

```

In [8]: 1 import pandas as pd
2 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
3
4 print("Original columns:", df.columns.tolist())
5
6 # Select correct columns explicitly
7 df_clean = df[[
8     'Country name',
9     'Ladder score',
10    'Logged GDP per capita',
11    'Social support',
12    'Healthy life expectancy',
13    'Freedom to make life choices',
14    'Generosity',
15    'Perceptions of corruption'
16 ]].copy()
17
18 # Rename properly
19 df_clean.columns = [
20     'Country',
21     'Happiness_Score',
22     'GDP',
23     'Social_Support',
24     'Life_Expectancy',
25     'Freedom',
26     'Generosity',
27     'Corruption'
28 ]
29
30 print(df_clean.head())

```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']

	Country	Happiness_Score	GDP	Social_Support	Life_Expectancy \
0	Finland	7.804	10.792	0.969	71.150
1	Denmark	7.586	10.962	0.954	71.250
2	Iceland	7.530	10.896	0.983	72.050
3	Israel	7.473	10.639	0.943	72.697
4	Netherlands	7.403	10.942	0.930	71.550

	Freedom	Generosity	Corruption
0	0.961	-0.019	0.182
1	0.934	0.134	0.196
2	0.936	0.211	0.668
3	0.809	-0.023	0.708
4	0.887	0.213	0.379

```
In [10]: 1 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
2 print("Original columns:", df.columns.tolist())
3
4 df_subset = df.iloc[:, :9] # Select only the first 9 columns
5 df_subset.columns = ['Country', 'Region', 'Happiness_Score', 'GDP', 'Social_Support',
6                      'Life_Expectancy', 'Freedom', 'Generosity', 'Corruption']
7
8 print(f"Dataset: {len(df_subset)} countries, {len(df_subset.columns)} columns")
9 print(df_subset.head())
```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']

Dataset: 137 countries, 9 columns

	Country	Region	Happiness_Score	GDP	Social_Support	\
0	Finland	7.804	0.036	7.875	7.733	
1	Denmark	7.586	0.041	7.667	7.506	
2	Iceland	7.530	0.049	7.625	7.434	
3	Israel	7.473	0.032	7.535	7.411	
4	Netherlands	7.403	0.029	7.460	7.346	

	Life_Expectancy	Freedom	Generosity	Corruption
0	10.792	0.969	71.150	0.961
1	10.962	0.954	71.250	0.934
2	10.896	0.983	72.050	0.936
3	10.639	0.943	72.697	0.809
4	10.942	0.930	71.550	0.887

```
In [11]: 1 region_map = {
2     'Finland': 'Western Europe',
3     'Denmark': 'Western Europe',
4     'Iceland': 'Western Europe',
5     'Netherlands': 'Western Europe',
6     'Sweden': 'Western Europe',
7     'Norway': 'Western Europe',
8     'Germany': 'Western Europe',
9     'France': 'Western Europe',
10    'Spain': 'Western Europe',
11
12    'India': 'South Asia',
13    'Pakistan': 'South Asia',
14    'Bangladesh': 'South Asia',
15
16    'China': 'East Asia',
17    'Japan': 'East Asia',
18    'South Korea': 'East Asia',
19
20    'Nigeria': 'Sub-Saharan Africa',
21    'Kenya': 'Sub-Saharan Africa',
22    'Ethiopia': 'Sub-Saharan Africa',
23    'Congo (Kinshasa)': 'Sub-Saharan Africa',
24    'Sierra Leone': 'Sub-Saharan Africa',
25
26    'Brazil': 'Latin America',
27    'Argentina': 'Latin America',
28    'Mexico': 'Latin America'
29 }
30
31 df_clean['Region'] = df_clean['Country'].map(region_map)
32 df_clean = df_clean.dropna(subset=['Region'])
```

In [7]:

```
1 import pandas as pd
2 import plotly.express as px
3 import plotly.io as pio
4 pio.renderers.default = 'browser'
5 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
6 df_clean = df[['Country name', 'Ladder score']].copy()
7 df_clean.columns = ['Country', 'Happiness_Score']
8 region_map = {
9     'Finland': 'Western Europe',
10    'Denmark': 'Western Europe',
11    'Iceland': 'Western Europe',
12    'Netherlands': 'Western Europe',
13    'Sweden': 'Western Europe',
14    'Norway': 'Western Europe',
15    'Germany': 'Western Europe',
16    'France': 'Western Europe',
17    'Spain': 'Western Europe',
18
19    'India': 'South Asia',
20    'Pakistan': 'South Asia',
21    'Bangladesh': 'South Asia',
22
23    'China': 'East Asia',
24    'Japan': 'East Asia',
25    'South Korea': 'East Asia',
26
27    'Nigeria': 'Sub-Saharan Africa',
28    'Kenya': 'Sub-Saharan Africa',
29    'Ethiopia': 'Sub-Saharan Africa',
30    'Congo (Kinshasa)': 'Sub-Saharan Africa',
31    'Sierra Leone': 'Sub-Saharan Africa',
32
33    'Brazil': 'Latin America',
34    'Argentina': 'Latin America',
35    'Mexico': 'Latin America'
36 }
37 df_clean['Region'] = df_clean['Country'].map(region_map)
38 df_clean['Region'] = df_clean['Region'].fillna('Other')
39 print("Columns:", df_clean.columns)
40 print(df_clean.head())
41 print(df_clean['Region'].value_counts())
42 region_avg = df_clean.groupby('Region')['Happiness_Score'].mean().reset_index()
43 region_avg = region_avg.sort_values('Happiness_Score')
44
45 print(region_avg) # debug
46 fig1 = px.bar(
47     region_avg,
48     x='Happiness_Score',
49     y='Region',
50     orientation='h',
51     color='Region'
52 )
53
54 fig1.update_xaxes(range=[0, region_avg['Happiness_Score'].max()])
55
56 fig1.update_layout(
57     title="Western Europe Shows the Highest Happiness Levels Among Selected Regions",
58     plot_bgcolor='white'
59 )
60 fig1.show()
```

```
Columns: Index(['Country', 'Happiness_Score', 'Region'], dtype='object')
```

	Country	Happiness_Score	Region
0	Finland	7.804	Western Europe
1	Denmark	7.586	Western Europe
2	Iceland	7.530	Western Europe
3	Israel	7.473	Other
4	Netherlands	7.403	Western Europe
	Other	114	
	Western Europe	9	
	Sub-Saharan Africa	5	
	Latin America	3	
	East Asia	3	
	South Asia	3	

```
Name: Region, dtype: int64
```

	Region	Happiness_Score
4	Sub-Saharan Africa	3.980800
3	South Asia	4.291000
2	Other	5.480491
0	East Asia	5.966000
1	Latin America	6.159667
5	Western Europe	7.224667

```
In [15]: 1 pip install plotly
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: plotly in c:\programdata\anaconda3\lib\site-packages (5.9.0)
Requirement already satisfied: tenacity>=6.2.0 in c:\programdata\anaconda3\lib\site-packages (from plotly) (8.0.1)
Note: you may need to restart the kernel to use updated packages.
```

In [11]:

```
1 import pandas as pd
2 import plotly.express as px
3 import plotly.io as pio
4
5 pio.renderers.default = 'browser' # or 'notebook' if you prefer
6
7
8
9 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
10
11 df_clean = df[[
12     'Country name',
13     'Ladder score'
14 ]].copy()
15
16 df_clean.columns = ['Country', 'Happiness_Score']
17
18 print("Data preview:")
19 print(df_clean.head())
20
21
22 top8 = df_clean.nlargest(8, 'Happiness_Score').copy()
23 top8['Group'] = 'Top 8'
24
25 bottom8 = df_clean.nsmallest(8, 'Happiness_Score').copy()
26 bottom8['Group'] = 'Bottom 8'
27
28
29 combined = pd.concat([bottom8, top8])
30 combined = combined.sort_values('Happiness_Score')
31
32
33 global_avg = df_clean['Happiness_Score'].mean()
34
35
36 fig = px.bar(
37     combined,
38     x='Happiness_Score',
39     y='Country',
40     orientation='h',
41     color='Group',
42     color_discrete_map={
43         'Top 8': 'red',
44         'Bottom 8': 'blue'
45     }
46 )
47
48
49 fig.add_vline(x=global_avg, line_dash="dash")
50
51
52 fig.update_xaxes(range=[0, combined['Happiness_Score'].max()])
53
54 fig.update_layout(
55     title="Top Countries Are Nearly Twice as Happy as the Lowest Ranked Ones",
56     plot_bgcolor='white'
57 )
58
59
60 fig.show()
```

Data preview:

	Country	Happiness_Score
0	Finland	7.804
1	Denmark	7.586
2	Iceland	7.530
3	Israel	7.473
4	Netherlands	7.403

In [10]:

```
1 import pandas as pd
2 import plotly.graph_objects as go
3 import plotly.io as pio
4 pio.renderers.default = 'browser'
5 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
6 df_clean = df[[
7     'Country name',
8     'Ladder score',
9     'Logged GDP per capita',
10    'Freedom to make life choices'
11]].copy()
12 df_clean.columns = [
13    'Country',
14    'Happiness_Score',
15    'GDP',
16    'Freedom'
17]
18 region_map = {
19    'Finland': 'Western Europe',
20    'Denmark': 'Western Europe',
21    'Iceland': 'Western Europe',
22    'Netherlands': 'Western Europe',
23    'Sweden': 'Western Europe',
24    'Norway': 'Western Europe',
25    'Germany': 'Western Europe',
26    'France': 'Western Europe',
27    'Spain': 'Western Europe',
28
29    'India': 'South Asia',
30    'Pakistan': 'South Asia',
31    'Bangladesh': 'South Asia',
32
33    'China': 'East Asia',
34    'Japan': 'East Asia',
35    'South Korea': 'East Asia',
36
37    'Nigeria': 'Sub-Saharan Africa',
38    'Kenya': 'Sub-Saharan Africa',
39    'Ethiopia': 'Sub-Saharan Africa',
40    'Congo (Kinshasa)': 'Sub-Saharan Africa',
41    'Sierra Leone': 'Sub-Saharan Africa',
42
43    'Brazil': 'Latin America',
44    'Argentina': 'Latin America',
45    'Mexico': 'Latin America'
46}
47 df_clean['Region'] = df_clean['Country'].map(region_map)
48 df_clean['Region'] = df_clean['Region'].fillna('Other')
49 regions = [
50    'Western Europe',
51    'Latin America',
52    'East Asia',
53    'Sub-Saharan Africa',
54    'South Asia'
55]
56 subset = df_clean[df_clean['Region'].isin(regions)]
57 grouped = subset.groupby('Region')[['GDP', 'Freedom']].mean().reset_index()
58 print(grouped)
59 fig3 = go.Figure()
60 fig3.add_trace(go.Bar(
61     x=grouped['Region'],
62     y=grouped['GDP'],
63     name='GDP'
64 ))
65 fig3.add_trace(go.Bar(
66     x=grouped['Region'],
67     y=grouped['Freedom'],
68     name='Freedom'
69 ))
70 fig3.update_layout(
```

```
71     barmode='group',
72     title="Western Europe Leads in Both Economic Strength and Personal Freedom Compared",
73     xaxis_title="Region",
74     yaxis_title="Average Value",
75     plot_bgcolor='white'
76 )
77
78 fig3.show()
```

	Region	GDP	Freedom
0	East Asia	10.349000	0.799333
1	Latin America	9.797000	0.826667
2	South Asia	8.661333	0.836000
3	Sub-Saharan Africa	7.818800	0.691600
4	Western Europe	10.853667	0.895556

In []:

1